Performing Social Closure: How College Board Geomarkets Structure Recruiting Visits by Selective Colleges

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ABSTRACT

Selective colleges are sites of social closure. Scholarship examines how selective colleges evaluate applicants, but colleges do not passively accept applications. They recruit. If college recruiting territories are structured by shared infrastructures, then mechanisms of social closure operate upstream of admissions. This article examines how College Board Geomarkets and the Market Segment Model structure high school recruiting visits by selective colleges. Geomarkets carve metropolitan areas into smaller geographic units meant to define local recruiting territories. The Market Segment Model predicts how student demand varies by Geomarket as a function of social class. These market devices became embedded in organizational structures and in third-party products that produce college recruiting behavior. Using data on recruiting visits made in 2017 by 42 selective colleges, we find that patterns of recruiting behavior are consistent with the Market Segment Model and that Geomarkets explain substantial variation in which high schools receive visits, even after controlling for school- and neighborhood-level covariates. The market devices that college admissions offices incorporate constitute a shared infrastructure that performs social closure – matching a socioeconomic hierarchy of communities to a socioeconomic hierarchy of colleges – prior to the evaluation of applications. We introduce the concept of infrastructural collusion: shared reliance by direct competitors on market devices that encode class hierarchy into the geography of opportunity.

1 Introduction

> The hierarchical structure of collegiate competition largely reflects the stratified social and economic dimensions of the communities from which colleges draw their students. Competition among colleges, as admissions officers have told us for so long, is in fact, a matter of keeping company with one's peers (Zemsky and Oedel 1983)

Selective colleges are sites of social closure. They confer access to scarce credentials and networks that facilitate entry into privileged occupations (Chetty et al. 2023), while enrolling student bodies that remain sharply stratified by class and race (Chetty et al. 2020a; Abramitzky et al. 2024). Zemsky and Oedel (1983) observes that competition between colleges occurs within "collective markets," defined by socioeconomic homophily, whereby like-colleges compete for like students on the basis of class. The Consortium on Financing Higher Education (COFHE) – a group of selective, private colleges – is one such collective market. "What every envious admissions officer outside the Consortium understands," Zemsky and Oedel (1983, 45) write, "is that these thirty institutions are strengthened by their very collectivity, though they are in fact one another's principal competitors." With support from College Board, Zemsky developed these observations into market devices for colleges that intervene on the college choice process of students.

Existing explanations of social closure emphasize exclusionary strategies of status groups that redefine admissions criteria in ways that benefit the privileged group (Karabel 2005, 1984) and then diffuse to other colleges [CITE BROADER LITERATURE]. Another literature traces legal and policy debates over race-conscious admissions (Warikoo 2025). These explanations center the admissions process. But colleges do not passively accept applications. They recruit (Stevens 2007). The markets they recruit from are not given, they are constructed. If college recruiting territories are structured by shared recruiting infrastructures, then the mechanisms of social closure operate upstream of admissions, raising questions about how recruiting territories are defined and made actionable for colleges.

College recruiting efforts depend on third-party "market devices" that classify high schools and communities on behalf of colleges looking for students. Market devices are tools that calculate and classify. They become incorporated into organizational goals and routines in ways that produce a menu of possible and likely actions. Market devices engender classification situations Fourcade and Healy (2013)], whereby a vertical hierarchy of products is matched to a vertical hierarchy of consumers. We argue that Geomarkets and the Market Segment Model – developed by Zemsky with support from College Board – are market devices that match "collective markets" of selective colleges to affluent communities, performing social closure upstream of admissions.

Geomarkes carve metropolitan areas into smaller geographic units meant to define local recruiting territories. The Market Segment Model categorized students into four market segments that are a function of class. Zemsky and Oedel (1983) argue that selective colleges primarily enroll students from high-income, college-educated households, but the relative abundance of these students differs across Geomarkets. Over time, Geomarkets and the Market Segment Model became embedded in college organizational structures and in shared infrastructure colleges draw from. Because affluent Geomarkets concentrate the students selective colleges are looking for, this infrastructure systematically directs recruiting attention toward wealthy communities.

This study examines how Geomarkets and the Market Segment Model structure high school recruiting visits by a sample of 42 colleges – 14 selective private research universities, 12 selective private liberal arts colleges, and 16 public research universities. While selective private colleges are understood as instruments of intergenerational class transmission, public research universities have been engines of social mobility. However, state disenvestment encourages public research universities to search for nonresident tuition revenue in the same set of affluent Geomarkets as their private peers.

Analyses are guided by three research questions. First, to what extent are visit patterns consistent with recommendations from the Market Segment Model? To what extent do Geomarkets explain which high schools receive recruiting visits? How does Geomarket popularity – measured as number of visits per high school – mediate the relationship between school socioeconomic and racial/ethnic composition and the probability of a school receiving a visit?

For the selective private – and most public – colleges in our sample, the Market Segment Model recommends focusing on affluent Geomarkets located in regions where students are likely to consider a college like yours (Zemsky and Oedel 1983). Although a small number of (mostly public) colleges follow an in-state recruiting strategy, most colleges in our sample follow a national recruiting strategy, in which the majority of visits are to schools outside their home region (e.g., New England). Visits by private colleges – and out-of-state visits by public universities – are highly concentrated in affluent Geomarkets. With respect to RQ2, fixed effects regression models find that Geomarkets explain substantial variation in recruiting visits even after controlling for a strong set of school- and neighbohood-level controls. With respect to RQ3, regression models find that schools with a low share of free/reduced lunch students benefit more from being located in a popular geomarket than schools with a high share of free/reduced lunch and schools with a modest share of Black/Hispanic students benefit more from being located in a popular Geomarket than schools with a high share of Black/Hispanic enrollment. Drawing from Fourcade and Healy (2017), a wealthy, preominantly white school in a wealthy, predominantly white Geomarket is a "good match" with what selective colleges expect.

We make targeted contributions to the sociology of education and a broader contribution to the mechanisms of social closure. Scholarship on selective college access focuses on conflict over admissions criteria and who benefits [CITE], while scholarship on enrollment management shows that colleges devote substantial resources identifying and recruiting applicants [CITE]. Both literatures center colleges as protagonists, ignoring business-facing intermediaries that classify students on behalf of colleges. Our research points to an upstream ecosystem of third-party vendors, products, and professional associations that that collectively construct the infrastructure that colleges utilize to plan and implement recruiting strategy. Geomarkets and the Market Segment Model were grafted atop the history of racial and class segregation and became embedded within third-party products and college organizational structures. These market devices contribute to social closure by facilitating a tiered matching of colleges to places, encouraging selective private colleges to target high-income localities nationwide, while working class communities are left to less selective institutions. Although public research universities were engines of social mobility in the era of strong state funding, the desire to increase nonresident enrollment encourages to encourages public research universities to target the same set of affluent Geomarkets coveted by selective private institutions.

The manuscript is organized as follows. First, we review scholarship on college access and enrollment management. Second, we describe the case of Geomarkets and the Market Segment Model. Third, we develop a theoretical framework based on market devices and classification situations. Fourth, we describe data and methods. Fifth, we present results. Finally, we discuss implications for scholarship. Scholarship at the intersection of quantification and education has considered consumer-facing intermediaries (e.g., college rankings) that discipline consumers and, in turn, businesses [CITE]. However, intermediaries that classify consumers on behalf of businesses are more pervasive and more sophisticated (Fourcade and Healy 2024). In the education industry, these intemediaries match ordinally arranged consumers to ordinally arranged products. We argue that sociology of education can develop novel insights by shifting the object of inquiry from students and direct providers education (schools, and colleges) to the market devices that classify students for direct providers.

2 Literature Review: College Access in the Enrollment Economy

Social closure, associated with Max Weber [CITE], is the process by which advantaged groups maintain their privilege by restricting access to resources and opportunities that confer economic and social privilege. Weber conceived of social closure as competition between status groups, defined of "associational communities comprised of all who share a sense of status equality" (Karabel 1984, 3) based on membership of a characteristic (e.g., racial, ethnic, religious, social) that may be the basis of exclusion. Following the transition to rational-legal authority defined by universalistic bureacracies, organizations that confer economic and social privileges become the site of contestation between dominant groups seeking exclusionary closure and marginalized groups seeking usurptionary closure. Colleges are the exemplar site of contestation (Collins 1979). Weber states,

> The development of the diploma from universities …make for the formation of a privileged stratum in bureaus and in offices. Such certificates support their holders'…claims to monopolize socially and economically advantageous positions. When we hear from all sides the demand for an introduction of regular curricula and special examinations, the reason behind it is, of course, not a suddenly awakened 'thirst for education' but the desire for restricting the supply for these positions and their monopolization by the owners of educational certificates…As the acquisition of the educational certificate requires considerable expense and a period of waiting for full remuneration, this striving means a setback for talent (charisma) in favor of property.

In the U.S., selective colleges confer access to privileged social and economic positions. As college access expands to previously excluded populations, scholarship in sociology and economics document growth in the benefits of graduating from a selective institution (Chetty et al. 2023; Gerber and Cheung 2008), and growth in competition for access to selective institutions [CITE]. Based on data from the IRS, the "Mobility Report Cards" (Chetty et al. 2020b) measures both parental household income and student earnings post-graduation at the college level for students who were born in 1991 and were freshman around 2009. Although low-income students students who graduate from selective colleges have strong labor market outcomes (Chetty et al. 2020a), access to selective institutions is dominated by weealthy households. For example, at Colorado College, one of the colleges in our analysis sample, median family income was about $379,000 (2026 CPI), with just 2% of freshman coming from the bottom income quintile, 78% of freshman coming from the top quintile, 54% of freshman from the top 5%, and 24% of freshman from the top 1% (Chetty et al. 2020b).

Several literatures in the sociology of education have analyzed the admissions criteria utilized by selective colleges and which applicants these criteria benefit. The Weberian status group social closure tradition is articulated by the historical sociology of (Karabel 2005, 1984). Leaders of Ivy League institutions were themselves members of the elite Protestant status group and had an interest in blocking Jewish enrollment growth by developing "character" admissions criteria associated with distinctly upper-class Protestant activities and traits. [TRANSITION?] A robust literature analyzes the rise and fall of affirmative action (Hirschman et al. 2016; Grodsky 2007; Warikoo 2025; Karen 1991). Another literature ties the growing importance of SAT/ACT scores in selective college admissions to the rise of college rankings systems that reward colleges for enrolling high-scoring students (Grodsky et al. 2008; Alon and Tienda 2007; Espeland and Sauder 2016). Recent scholarship examines the implementation of "test optional" policies (Hsu and Sharkey 2025) and holistic admissions processes that evaluate applicants with respect to the opportunities available in their local context [CITE BASTEDO]. Although these studies offer rich accounts of how colleges evaluate applicants, the scholarly preoccupation with selective admissions obscures the basic fact that overwhelming majority of colleges – including most selective colleges – cannot thrive by passively receiving applications.

The "enrollment economy" refers to reliance on student enrollment for organizational stability.

The concept was introduced by Clark (1956) [p. 332], in his analysis of California adult education institutions: "The most important pressures on these schools in their day-to-day administration arise from the *enrollment economy* …Financial support from the state is figured by the hours of attendance logged the previous year …and a major slump in attendance constitutes a serious threat to organizational welfare." (p. 332). At the opposite end of the prestige spectrum, Winston (1999, 17) describes dependence on students by selective private colleges: "the technology of producing much of what is sold in higher education is unusual in that colleges can buy important inputs to their production only from the customers who buy their products; that is, higher education uses a customer-input technology. Colleges depend on enrolled students for tuition revenue and on alumni for donation revenue. They depend on enrolled students for peer learning effects. Furthermore, the characteristics of enrolled students – for example, average SAT scores – substantially determine their national ranking, which then determines student demand for the subsequent cohort of prospects [CITE] and also affects research funding and donation revenue [CITE].

Organizational stability in the enrollment economy is the responsibility of enrollment management (EM). As a profession, EM integrates techniques from marketing and economics in order to "influence the characteristics and the size of enrolled student bodies" (Hossler and Bean 1990, xiv). As an administrative structure, the "office of enrollment management" typically controls and coordinate the functional activities of admissions, financial aid, and recruiting (Kraatz et al. 2010; Hossler and Bontrager 2014).[1] The administrative structure of EM offers one means of categorizing scholarship about the supply-side of college access. Accross disciplines, the majority of scholarship focuses on admissions (e.g., Hirschman et al. 2016; Taylor et al. 2024; Posselt 2016; Killgore 2009) or on financial aid (e.g., Hurwitz 2012; Leeds and DesJardins 2015). These literatures offers less insight into how colleges identify and cultivate prospective applicants before admissions evaluation occurs. Therefore, a growing literature in sociology analyzes the earlier "recruiting" stages of acquiring "leads," and soliciting "inquiries" and applications [CITE]. Several monographs analyze recruiting as part of broader studies on enrollment management (Stevens 2007), high school to college transitions (Holland 2019), elite reproduction (Khan 2011), or political economy (Cottom 2017). From the

[1] Kraatz et al. (2010) exemplifies a literature at the intersection of EM and organizational behavior that shows how concerns about the enrollment economy motivate changes in organizational behavior, such as diversifying degree programs or changing the name of the organization (Brint and Karabel 1989; Kraatz and Zajac 1996; Jaquette 2013)

student perspective, Holland (2019) finds that underrepresented students were drawn to colleges that made them feel wanted, often attending institutions with lower graduation rates and requiring larger loans than other college options. Cottom (2017) shows that for-profit colleges found a niche in Black and Latinx communities because traditional colleges ignored these communities.

Stevens (2007) provides an ethnography of enrollment management at a selective private liberal arts college. Being tuition-dependent and sensitive about its position in the rankings, The College focuses on enrollment goals related to academic profile and revenue generation. Stevens (2007) highlights the relational function of off-campus recruiting visits: "the College's reputation and the quality of its applicant pool are dependent upon its connections with high schools nationwide" (p. 54). During the autumn "travel season," admissions officers visit selected high schools across the country "to spread word of the institution and maintain relationships with guidance counselors" (p. 53-54). Ruffalo Noel Levitz (2020) reported that "calling high school counselors after visits" was an effective and widely-used outreach strategy for both private universities and public universities. Thus, relationship maintenance remains a high-touch process even in the digital age. The College tended to visit the same "feeder" schools year after year because recruiting depends on long-term relationships. Visited schools tend to be affluent schools – in particular, private schools – that possess the resources to host a visit and enroll high-achieving students who can afford tuition.

Whereas scholars in the sociology of education have investigated recruiting as part of a broader analysis, AUTHORS (2021; 2022; 2024) have attempted to develop a literature that makes college recruiting behavior the object of empirical analysis. Salazar et al. (2021) argue that off-campus recruiting visits are a credible indicator of college enrollment priorities.[2] [INSERT SENTENCES SAYING HOW MUCH IS SPENT ON OFF-CAMPUS RECRUITING AND SCHOLARSHIP ON THE EFFECTS OF OFF-CAMPUS RECRUITING]. Salazar et al. (2021) analyzed recruiting visits made by public research universities. The authors found that 12 of the 15 universities made more visits to out-of-state schools than in-state schools. These out-of-state visits focused on affluent, predominantly white public high schools. Set against scholarship about nonresident enrollment growth as a response to declining state support (Jaquette and Curs 2015; Jaquette et al. 2016), these

[2]Sociological theories of organizational behavior argue that organizations expend substantial resources on goals they care about while symbolically adopting – highly visible, ceremonial behavior with few resources expending – goals demanded by stakeholders in the broader institutional environment [CITE].

results suggest that many public research universities prioritize enrollment from wealthy out-of-state students.

Jaquette et al. (2024) conduct a social network analysis of recruiting visits to private high schools by a sample of public research universities and selective private research universities. A visit indicates a social relationship between a college and a high school, suggesting that the college wants to enroll some students from the high school and the high school wants to send some students to the college. Both public and private universities made a disproportionately large number of visits to private schools relative to public schools. Analyses revealed high overlap in the recruiting networks of public and private universities. Salazar (2022) conducted geospatial analyses of out-of-state recruiting visits to Los Angeles and Dallas by four public research universities. She found that universities engage in "recruitment redlining – the circuitous avoidance of predominantly Black and Latinx communities along recruiting visit paths" (p. 585).

Upon review, scholarship locates inequality in college access in the behavior of households, schools, and colleges [CITE]. Scholarship on the supply-side of college access – selective admissions, EM, recruiting – argues that students are sorted by the actions of colleges, which are conditioned by macro-institutions in the organizational field [CITE]. This focus on college behavior understates the role of third-parties that mediate between consumers and sellers. Sociology of education has investigated intermediaries that "occupy the interface between consumers and producers" (Sharkey et al. 2023, 1), including scholarship on college rankings (Espeland and Sauder 2016) and school rankings (McArthur and Reeves 2022; Thompson 2024; Hasan and Kumar 2019). To our knowledge, intermediaries that classify consumers on behalf of sellers have rarely been the object of empirical research (but see Hamilton et al. 2024). Although consumer-facing intermediaries are salient to media outlets and researchers, intermediaries that classify consumers for businesses are more sophisticated, pervasive (McKelvey 2022; Muniesa et al. 2007; Cochoy 2007; Fourcade and Healy 2017), and may be highly salient to professionals responsible for recruiting customers. Scholarship on college access has ignored the infrastructural mechanism of third-party enrollment management tools that classify students and localities on behalf of colleges. These tools – which are built atop the foundation of segregation in American communities – facilitate a tiered matching of colleges to places by encouraging selective private colleges and public research universities target high-income localities nationwide, while encouraging less selective institutions to target nearby working- and middle-class communities.

Geomarkets were conceived in service of a class-based model of college choice (Zemsky and Oedel 1983). Subsequently, Geomarkets became the basis of College Board Enrollment Planning Service (EPS) software that helped colleges plan recruiting efforts and an organizing principle allocating admissions personnel to recruiting territories. We conceptualize these tools as "market devices" (Callon 1998) that create "classification situations" (Fourcade and Healy 2013), whereby an ordinal hierarchy of students is matched to an ordinal hierarchy of postsecondary institutions. Performativity is the idea that market devices are used in ways that create the behaviors they predict (MacKenzie and Millo 2003). Empirically, we show that Geomarkets explain substantial variation of high school visits by a sample of selective private colleges and public research universities, even after controlling for a strong set of school- and neighborhood-level covariates. Finally, we contribute to a critical sociology of the College Board. Existing scholarship focuses on College Board's consumer-facing testing products [CITE], mirroring the blind-spot observed in scholarship on intermediaries. This manuscript shows that College Board testing products are the basis for business-facing products that classify prospective students on behalf of colleges.

3 The Case: Geomarkets and the Market Segment Model

Geomarkets were conceived in the late 1970s, when colleges were concerned about declining demographics. Zemsky and Oedel (1983) [p. ix] write that demographer "Stephen Dresch (1975) had already frightened everyone who would listen with his projection of 40-percent declines in college enrollments by the close of the 1980s." Elite colleges responded by leaning on the associational configuration (Stevens and Gebre-Medhin 2016) to solve problems common to the group. The project began when UPenn president, Martin Meyerson, tasked UPenn professor Robert Zemsky, to figure out, " 'Who thinks about Penn?' " and " 'What other institutions do they think about when they think about us?' " (Zemsky and Oedel 1983, x):

> "In pursuit of these questions, we began working in 1978 with the Market Research Committee of the Consortium on Financing Financing Higher Education (COFHE).

> The Consortium, a group of thirty selective private institutions, had in its brief history developed a remarkable tradition of sharing data...For nearly a year, the Consortium's Market Research Committee provided focus to our efforts. Much of this project's framework – the Market Segment Model and its derivation from admissions officers' folklore was shaped and then reshaped in our discussions with members of the committee."

Zemsky and Oedel (1983) conceived of the Market Segment Model as an effort to "capture and quantify" (p. 11) the knowledge of admissions officers: "We believe that the intuitions of admissions officers actually comprise a remarkably systematic body of knowledge about the college selection process...Our research is based upon listening carefully to what admissions officers have to say (pp. 9-10). The"initial [analytic] task" was to define geographic boundaries "in a manner consistent with admissions officers' intuitive understanding of student pools" (Zemsky and Oedel 1983, 4). Quoting an admissions officer, Zemsky and Oedel (1983) [p. 11] write, " 'There are only three kinds of college-bound students: those who want to live at home, those who want to live on campus but bring their laundry home, and those who want to go far enough from home that Mom and Dad can't visit without calling first.' " As such, Zemsky and Oedel (1983, 11) created "three types of boundaries – region, state, and community." The initial regions in the project were New England, Middle States, and the South. Next, they "divided each state into as few as two and as many as thirty community-based enrollment markets or pools, for a total of 143 separate markets" (p. 11). These enrollment markets, later called Geomarkets, were intended to be consistent with the conception of a catchment market from the perspective of admissions counselors. Zemsky and Oedel (1983, 11–12) described the creation of Geomarket borders only briefly:

> In many cases, the market boundaries match formal political and educational divisions, reflecting natural channels of communication. Each major metropolitan area is composed of several markets, usually corresponding to the inner city, a first ring of suburbs, and an outer ring of suburbs. In more sparsely populated areas, communities are sometimes combined in order to make the analysis meaningful.

Over time, the project goal morphed into predicting student demand for any college from any locality. However, "to gain a truly comprehensive view of the collegiate enrollment market, we needed a database that described most institutions and most students" (Zemsky and Oedel 1983, x). The group approached College Board. "Coincidentally, the Board was reviewing its own efforts to help colleges estimate their enrollment potential" (Zemsky and Oedel 1983, x). College Board provided score-sending data for all 1980 SAT test-takers. For a given student, each college that receives a score from the student can be defined as "local" (college located in the the same Geomarket as the student), "in-state" (same state but different Geomarket), "regional" (same region but different state), or "national" (different region). Zemsky and Oedel (1983) classified each SAT test-taker into one of four student *market segments*. A student is categorized in the "local" market segment if they submit more SAT scores to local institutions than they do to in-state, regional, or national institutions. An "in-state" student submits more SAT scores to in-state institutions than they do to local, regional, or national institutions, etc. The Market Segment Model "was nothing more than a set of simple rules for disaggregating high school seniors into similar groups" (Zemsky and Oedel 1983, 4). "The model worked because students, once so disaggregated, appeared to behave in remarkably consistent ways."

Zemsky and Oedel (1983, 3) investigates which student characteristics predict whether a student belongs to the local, in-state, regional, or national market segment. The analyses identify four variables – educational aspirations, parental education, scholastic aptitude, and family income – that predict score-sending behavior, both individually and in combination. The resulting thesis is that student demand for college is a function of social origin. These four variables "reflect the basic social patterns of the nation; it would have been surprising if these were not the four social variables that best explained the patterns of college choice" (Zemsky and Oedel 1983, 33). The analyses conclude (p. 42) that "our research has simply demonstrated what everyone has always known: communities with high levels of family income and parental education are also communities in which students have higher than average SATs and more far-reaching aspirations."

However, Geomarkets differ in the relative abundance of students with particular socioeconomic characteristics. As college leaders sought new sources of patronage, the Market Segment Model yielded practical implications about which communities to target. Zemsky and Oedel (1983, 44) recommend that colleges target Geomarkets with desirable compositions of socioeconomic characteristics in order to reach students from desired student market segments:

> On occasion, senior spokespersons for the profession worry that students outside the

> main [Geo]market areas remain forgotten and hence, unchallenged. Inevitably, the increasing competition for students, the expense of travel and mailings, and internal political constraints compel institutions to concentrate their efforts where they will do the most good. The result is a natural reinforcing of the basic socioeconomic patterns that gave shape in the first place to the structure of college choice.

Zemsky and Oedel (1983) developed two analytic outputs, (1) the Market Segment Profile and (2) the Institutional Profile. The Market Segment profile showed – separately by Geomarket – the number of students from each market segment (local, in-state, regional, national), their socioeconomic characteristics, and average test scores. The Institutional Profile showed – separately by Geomarket and institution (your own or a competitor) – how many students from each market segment sent scores to that institution.

In 1984, College Board transformed the Market Segment Model and associated outputs into the Enrollment Planning Service (EPS), an early software-as-service platform sold to colleges that was based on Zemsky and Oedel (1983). For any Geomarket, colleges could obtain the Market Segment Report and the Institutional Profile for themselves or a competitor. Based on background conversations with EM professionals, EPS software also provided information about the score-sending behavior of individual high schools within each Geomarket. Typical College Board (2005) marketing material describes EPS as, "the marketing software that pinpoints the schools and Geomarkets where your best prospects are most likely to be found. With the click of a mouse, EPS provides you with comprehensive reports on your markets, your position in those markets, and your competition." Following Zemsky and Oedel (1983), the EPS platform encouraged users to, first, identify promising Geomarkets and, second, to identify desirable high schools for recruiting visits. To find new markets, EPS encouraged colleges to select peer colleges and identify Geomarkets where a large number of students sent scores to those colleges. Market research suggests that EPS was highly salient. Noel-Levitz (1998) reports that in 1995, 37% of 4-year publics and 49% of 4-year privates used EPS, while 41% of 4-year publics and 16% of 4-year privates used ACT's market analysis service product.

4 Theory: Market Devices and Classification Situations

The market devices literature examines how the components of markets are assembled and enacted, and how these processes shape behavior (Callon 1998; Muniesa et al. 2007; MacKenzie and Millo 2003). Market devices are discreet calculative tools (e.g., algorithms, metrics, graphs, rankings models, spreadsheets) that are incorporated into products or economic transactions. Muniesa et al. (2007, 2) define market devices as a "way of referring to the material and discursive assemblages that intervene in the construction of markets." Here, material refers to tangible tools and technologies, like spreadsheets, algorithms or software, while discursive refers to representational elements, such as categorical, quantitative, and ordinal variables. Assemblages refers to a bundle of heterogeneous components that work together, suggesting that market devices consist of and build upon other market devices. Geomarkets are a market device – based on the SAT – that helps enrollment managers define recruiting territories and make decisions about which high schools to recruit from. Market devices are non-neutral, actively shaping outcomes. Therefore, markets do not exist a priori waiting to be observed; they are constructed by market devices. From the market devices perspective, the transformation of American higher education "from a collection of local autarkies to a nationally integrated market" (Hoxby 1997, 1) can be (partially) explained as the consequence of enrollment managers using Geomarkets to achieve enrollment goals.

Theories (e.g., pricing theory, status attainment theory), by themselves, are not market devices. They are abstract ideas that can be applied to specific contexts. For example, homophily is the idea that people who share similar characteristics are more likely to connect (McPherson et al. 2001). Theories become part of market devices when components of the theory are embedded within the device. Zemsky and Oedel (1983) argue that homophily is the organizing principle of competition between colleges; similar colleges compete with one another for similar students. An analytic output from Zemsky and Oedel (1983) is the Institutional Profile, which provided information about students from a particular Geomarket who sent SAT scores to a particular institution. This functionality was subsequently embedded in EPS software, which encouraged colleges to investigate which Geomarkets have high demand for peer colleges

In the market devices framework, neither market devices nor actors have agency in isolation from one another. Whereas market devices are tools, "agencements" are "socio-technical configurations" (Çalışkan and Callon 2010) of market devices, people, and organizational arrangements (goals, roles, routines, incentives) designed to produce a menu of possible actions and to make some actions more likely than others. Agencement orients research to the intersection of what the market devices is capable of doing and the organizational context that is utilizing the device – in concert with other devices – to achieve organizational goals. Geomarkets are an input for enrollment management offices charged with realizing college enrollment goals. Geomarkets can be utilized to define recruiting territories and assign admissions officers to territories. EPS can be used to investigate which Geomarkets exhibit strong demand for the college, for peer institutions, and to plan which high schools in a Geomarket will be visited by which admissions officer.

Performativity is the idea that theories of market behavior do not merely describe behavior, rather market devices are constructed and acted upon in ways the bring about behavior consistent with the theory (Muniesa et al. 2007; MacKenzie and Millo 2003). For example, MacKenzie and Millo (2003) provide a historical sociology of the Chicago Board Options Exchange, which opened in 1973. An option is a contract that gives the holder the right to buy or sell an asset (e.g., a stock) at a given price. Economists Fischer Black and Myron Scholes developed the Black-Scholes model – a calculative market device – to predict the price of options Black and Scholes (1973), eventually winning the Nobel Prize in Economics. Initially, the Black-Scholes algorithm did not accurately predict the price of options. Black-Scholes model predictions became more accurate over time because options traders began embedding the logic and components of the model into spreadsheets/software. That is, the models predictions became more accurate because the model produced the prices it predicted (MacKenzie and Millo 2003)! Bastedo and Bowman (2010) provide an example of performativity for the case of US News College Rankings. The reputational assessment of senior administrators was a substantial component of the rankings algorithm. Bastedo and Bowman (2010) show that administrators based their assessment scores on the prior overall rankings. Therefore, the rankings algorithm disciplines administrators to provide assessments that are consistent with the prior rankings.

Although canonical contributions to the market devices literature did not focus on structural inequality – race, gender, class, Hirschman and Garbes (2019) argue that market devices literature can offer novel insights about the mechanisms of structural inequality. Analysis of the Houston residential real estate market by Korver-Glenn (2018, 2022) shows how market devices can be a mechanism of racial inequality. Every home listing is a assigned to a Houston Association of Realtors (HAR) "market area" – a market device akin to Geomarkets – based on the address of the property. In 70 of the 98 HAR market areas, one racial group comprised at least 50% of residents. Korver-Glenn (2022, 641) observed that "50 of these 70 market areas were majority-white – a disproportionately high number given that only 31 percent of the population within the encompassing area is white." On average, majority-white market areas contained about 20,000 residents, while majority-Black and majority-Hispanic market areas, respectively, contained 61,000 and 88,000 residents. Majority-white market areas also exhibited greater socioeconomic homogeneity. Because appraisals are based on recent comparable sales within the same market area (agencement), homes in high-income white neighborhoods received higher appraisal values than homes in high-income Black or Hispanic neighborhoods. College Board Geomarkets often exhibit similar differences in granularity. For example, the affluent, predominantly White "IL 9 - North Shore" Geomarket, had a 2020 population of about 161,000 while "IL 12 - South and Southwest Suburbs" Geomarket had a 2020 population of about 1.2 million.

The concept commensuration (Fourcade 2011; Espeland and Stevens 1998) helps us gain deeper insight about the agencement of market devices created by College Board. Commensuration is "the expression or measurement – of characteristics normally represented by different units – according to a common metric" (Espeland and Stevens 1998, 315). Commensuration is a process that requires work, akin to institutionalization. Once achieved, commensuration reduces the cognitive effort of decision-making. It is "a way to reduce and simplify disparate information into numbers that can easily be compared. This transformation allows people to quickly grasp, represent, and compare differences" (p. 316). For example, college rankings enable prospective students to compare colleges based on a single metric. Therefore, commensuration "can be understood as a system for discarding information and organizing what remains into new forms" (Espeland and Stevens 1998, 317). McArthur and Reeves (2022) analysis of UK school "league tables" – a commensurating market device that facilitated comparing schools on the basis of test scores – shows how commensuration can be a mechanism of social reproduction; professional/managerial households were more aware of these consumer-facing metrics and had resources to respond by moving to more expensive neighbhorhoods, near higher performing schools.

College Board Geomarkets are a commensurating market device that was built on the SAT and became an input into subsequent market devices. Drawing from Hoxby (2009), the SAT is a commensurating market device that enabled colleges to compare students from different schools, causing a "dramatic fall" in the cost of "colleges' information about students" (p. 102). The Market Segment Model is a theory of student demand that categorizes students into four market segments – local, in-state, regional, and national – based on SAT score-sending behavior. Geomarkets are a market device that defines geographic localities and enables colleges to commensurate these localities based on the number of students from each market segment. Whereas UK league tables allowed parents to compare geographically disparate schools based on test scores (McArthur and Reeves 2022), Geomarkets and the Market Segment Model enables enables colleges to compare geographically proximate and geographically disparate Geomarkets based on the number of students from each market segment. Each student market segment is associated with particular sets of institutions. Zemsky and Oedel (1983) states that selective colleges primarily draw students from the regional and national market segments. The key to finding new student markets, then, is identifying Geomarkets with large numbers of regional and national students. Geomarkets are then incorporated into the Enrollment Planning Service product, enabling colleges to decide which Geomarkets to recruit from and which high schools to visit (Takamiya 2005).

Classification situations. Scholarship on classification situations (Fourcade and Healy 2013, 2024, 2017; Fourcade 2016) helps conceptualize how Geomarkets faciliate social and racial stratification in college access. Traditionally, sociology defined social class from the production side of the economy (e.g., worker vs owner; clerical, manager, professional). By contrast, Fourcade and Healy (2013) argue that processes of ordinalization "endogenous to the market" (p. 560) define class through effects on the *consumption* side of the economy.

Classification situations are ordinal interventions whereby organizations "sort and slot people into new types of categories with different economic rewards or punishments attached to them" (Fourcade and Healy 2013, 561). The canonical example is credit scores (Poon 2007), which enable third-party intermediaries on supply side to "slice consumers into behaviorally-defined risk groups, and price offerings to them accordingly. On the demand side, consumers find themselves …fitting into these categories – which, by design, are not constructed from standard demographic classifications such as race" (Fourcade and Healy 2013, 560) but are highly correlated with social categories. The goal of classification situations is "good matches," whereby an ordinal hierarchy of products match to an ordinal hierarchy of people. Businesses "seek good matches when they want to link a wealthy customer with a quality credit card, or a heavy drinker with a bad insurance plan" (Fourcade and Healy 2017, 17). Whereas economics tends to view such matches favorably – for example, Hoxby (2009, 106) writes the "efficient outcome allocates students to colleges based on students' ability to benefit from the …magnitude of the human capital investment that the college offers" – Sociology tends to highlight the predatory, extractive inclusion of formerly excluded consumers (e.g., for-profit colleges) (Cottom 2020).

Geomarkets and the Market Segment Model function together as an integrated market device that matches a socioeconomic hierarchy of places to a socioeconomic hierarchy of colleges.[3] Zemsky and Oedel (1983) defined student market segments in ordinal terms – local, in-state, regional, and national – and found that student market segment was primarily a function of social class, particularly parental income and parental education. Zemsky and Oedel (1983) also conceived of colleges as an ordinal hierarchy. As in Fourcade and Healy (2017), good matches are the key to realizing enrollment goals. In turn, Zemsky and Oedel (1983) argue that good matches are defined by homophily, the idea that actors who share common characteristics are likely to form connections with one another (McPherson et al. 2001; Chun 2021). Zemsky and Oedel (1983) conduct a network analysis whereby two colleges are defined as competitors if a large number of students submit SAT scores to both colleges, Zemsky and Oedel (1983) "draw[ing] a fundamental conclusion about the structure of college choice: collegiate competition occurs principally between like institutions" (p. 46). Subsequent analyses investigate the tuition price and the socioeconomic composition of institutions in competition with one another. The takeaway is that like-colleges compete for like-students as defined by social class, with private selective colleges competing directly for students, charging the highest prices, and enrolling students with the highest socioeconomic

[3] Whereas Fourcade and Healy (2024) analyze devices that classify individuals, the Market Segment Model harkens back to the earlier era of geodemography, in which market research classified places for merchants based on the attributes of people who lived there (Leyshon and Thrift 1999; McKelvey 2022)

status (p. 72):

> Students describe themselves socially simply by telling us the colleges and universities in which they are interested. The layering of collegiate competition is primarily a socioeconomic layering. The hierarchical structure of collegiate competition largely reflects the stratified social and economic dimensions of the communities from which colleges draw their students. Competition among colleges, as admissions officers have told us for so long, is in fact, a matter of keeping company with one's peers.

4.1 Research questions and expected results

Our overarching research question is, what is the relationship between college Board Geomarkets and high school recruiting visits of selective private colleges and public research universities? Colleges engage in enrollment management behaviors designed to realize enrollment goals. Whereas Geomarkets, the Market Segment Model, and EPS are market devices and/or components of market devices, the patterns of recruiting visits from colleges to high schools describe the performativity consequences of colleges using these market devices to realize enrollment goals.

Our first research question asks, to what extent are visit patterns consistent with recommendations from the Market Segment Model? The Market Segment Model is an algorithm for identifying desirable recruiting territories. For selective private – and most public – institutions in our sample, the Market Segment Model recommends focusing on affluent Geomarkets located in regions where many students are likely to consider a college like yours (Zemsky and Oedel 1983).

The US postsecondary education system is highly ordinal and so are student market segments. According to Zemsky and Oedel (1983), selective private colleges primarily depend on students from the regional and national market segments. Further, regional and national students tend to come from affluent, college-educated households. Affluent households are not scattered randomly across the country, but rather clump together in particular geographic communities. Like UK "league tables" (McArthur and Reeves 2022), Geomarkets are a means of commensurating geographically disparate localities in terms of the number of students from each market segment. Imagine that the University of Arkansas has substantial regional but not national student demand and wants to enroll more out-of-state students. The Market Segment Model points to the affluent "TX 23 –

Collin & Rockwall" (near Dallas) and the "TX 15 – Northwest Houston & Conroe School District" Geomarkets, which contain large numbers of regional and national students, but not the poorer "TX 19 – City of Dallas" or "TX 17 – City of Houston (East)," which contain relatively few regional and national students. We answer RQ1 using simple descriptive statistics. We predict most colleges in our sample will adhere to a regional or national (but not an in-state) recruiting strategy, depending on whether they have regional or national enrollment demand. We also predict that colleges will disproportionately visit high schools located in affluent Geomarkets.

RQ2 asks, to what extent do College Board Geomarkets (X) explain which high schools receive recruiting visits (Y)? We argue that Geomarkets explain college recruiting visits to high schools because they have become embedded in the organizational and technological infrastructures that generate them. These embeddings exemplify agencement, whereby Geomarkets transform from a mere classificatory device into a durable "socio-technical arrangement" of software, data fields, reports, and organizational roles, making Geomarkets a default unit for planning, staffing, and evaluation of recruitment. Through this agencement, Geomarkets may have performative consequences for recruiting behavior – even after controlling for school- and neighborhood-level covariates – because once a classification scheme is built into workflows, behavior will tend to align with it. We provide three examples of embedding Geomarkets into recruiting infrastructure.

First, although Geomarkets were initially designed to approximate the territories of admissions officers (Zemsky and Oedel 1983), over time they became an organizing principle through which admissions offices define recruiting territories and assign them to officers. On background, admissions officers and EM consultants told us that Geomarkets became part of the shorthand lexicon of the National Association of College Admissions Officers (NACAC). Admissions officers might introduce themselves by saying, "I recruit 'PA 2'," referring to the "Chester County, PA" Geomarket. We find evidence for this claim in the LinkedIn profiles of admissions officers. Examples include: "Personally managed schools in Indiana geomarkets 1, 2, 3, 4, 5, 6, 7, & 9" [CITE]; "Led recruitment efforts in geomarket MA06, the greater metro Boston area" [CITE]; and "Increased applications from the Long Island geomarket NY 16 - NY 21 for the 2021 admission cycle by 18.3%" [CITE]. Institutional materials also incorporate Geomarkets; a report from the Dartmouth College Admissions Ambassador Program lists the number of interviewers and applicants by Geomarket [CITE]. Anecdotal evidence suggests that admissions offices allocate relatively more admissions officers to affluent Geomarkets.[4].

Second, similar to the Black-Scholes model being written into spreadsheets used by options traders (MacKenzie and Millo 2003), Geomarkets and the logic of the Market Segment Model became the basis of the widely used EPS software, which encouraged colleges to identify promising Geomarkets and then select high schools to visit within them. EPS promotional material from College Board (2022) includes a screenshot of the user-interface navigation menu, which includes tabs for "Plan Travel," "Research High Schools," and "Competitive Analysis." This agencement integrates Geomarkets and the Market Segment Model logic into software-based workflows of territory assignment and travel planning, shaping the menu of possible recruiting visits and making some visits more likely than others. On background, EM professionals told us that EPS software enabled colleges to plan travel without relying on admissions officers having strong local knowledge of their territories. EPS software often recommended visiting the same sets of affluent, high-achieving schools visited by other colleges. Some savvier, quantitatively adept offices, however, used EPS to visit schools that EPS recommended ignoring because there would be less competition for the good students who attended these schools.

Third, Geomarkets have been integrated into Customer Relationship Management (CRM) and information systems used by colleges. Slate, the market-leading admissions CRM, allows institutions to assign admissions recruiters and prospective students to Geomarkets [CITE]. The LinkedIn profile of an admissions officer states, "Utilized Slate to create territory management tools for the recruitment team. Generated reports broken down by state and geomarket and sorted by high school to reflect applicant/admit/deposit data for past 3 cycles" [CITE]. Similarly, Oracle's PeopleSoft Campus Solutions student information system product integrates Geomarkets (referred to as "EPS Market Codes") and also EPS software [CITE]. These examples illustrate market devices as assemblages of heterogeneous components that build on one another. Integrating Geomarkets within third-party administrative infrastructuresmakes it easier for colleges to incorporate Geomarkets into recruiting workflows, including decisions about which high schools to visit.

RQ3 asks, to what extent does Geomarket popularity – measured as the number of visits per school

[4] For example, the University of Chicago assigns one recruiter to the "IL 9 - North Shore" Geomarket, which had a 2020 population of about 161,000, and one recruiter to the "IL 12 - South and Southwest Suburbs" Geomarket, which had a 2020 population of about 1.2 million [CITE]

– mediate the relationship between school composition and the probability of a receiving a visit? Previous research shows that having a high proportion of free-reduced lunch or Black/Hispanic students is negatively associated with receiving a visit (Salazar et al. 2021). Therefore, RQ3 investigates whether high- versus low-SES schools (and low versus high share of Black/Hispanic enrollment) benefit more from being located in a popular Geomarket. We hypothesize that high-SES and low Black/Hispanic enrollment schools benefit more from being located in a popular Geomarket. Our rationale draws from Fourcade and Healy (2017), who highlight the role of "good matches" between ordinally classified products and ordinally classified customers. Fourcade (2016, 20–21) offers an anecdote:

> One of the authors had a credit card transaction denied while trying to refill her car's gas tank in a high-poverty Oakland neighborhood. The clerk indicated with resigned anger that the location of his gas station was the likely cause of the problem. Back in Berkeley a few minutes later, the card worked without a hitch.

Not coincidentally, Oakland is in Geomarket "CA 7 – City of Oakland" whereas Berkeley is in Geomarket "CA 8 – Alameda County excluding Oakland." Fourcade and Healy (2017, 17) write that " 'good matches' are perceived as natural because they fit well with how things already are." An affluent school in an affluent Geomarket is a good match because it exemplifies the schools we expect to find. We predict that such schools benefit by receiving more visits than an affluent school in a less affluent Geomarket. By contrast, a poor or predominantly Black/Hispanic school in an affluent Geomarket does not match expectations and will not reap the same locational benefits.

5 Methods

5.1 Data and Sample

Sample of colleges. This manuscript draws from a data collection that tracked off-campus recruiting visits made during the 2017 calendar year by a convenience sample of colleges. The data collection sample consisted of: all public research-extensive universities as defined by the 2000 Carnegie Classification (N=102), all private universities in the top 100 of the 20XX U.S. News and World Report National Universities rankings (N=57), and all private colleges in the top 50 of the 20XX U.S. News and World Report Liberal Arts Colleges rankings (N=47). The rationale for different criteria for inclusion across different university types is because these criteria reflected the interest of a foundation that funded the data collection.

For universities in our data collection sample, we investigated their admissions website for pages that posted their upcoming off-campus recruiting visits. For institutions that posted such pages, we scraped the pages once per week throughout 2017.[5] Many universities in our data collection sample only posted certain kinds of events (e.g., hotel receptions and national college fairs) but not others (e.g., day-time visits to high schools). These institutions were excluded from the analysis sample. The final analysis sample for this article consists of 16 public research universities, 14 private selective research universities, and 12 private seletive liberal arts colleges that we believed posted all varieties of off-campus recruiting events on their admissions website.[6]

Sample of high schools. The analysis sample of high schools consists of all public and private high schools that met the following criteria in the 2017-18 academic year: enrolls at least 10 12th grade students; is not a virtual school; is located in the 50 U.S. states, the District of Columbia, or land regulated by the Bureau of Indian Affairs. We also excluded "special education schools" (481 public, 314 private) and "alternative education schools" (2,002 public, 147 private) because these school categories received very few visits. Additionally, the public schools in these categories had higher rates of missing values for school-level measures of math and reading achievement (about 50% missing). The analysis sample consists of 18,326 public high schools (18,110 regular, 216 career and technical) and 3979 private high schools (3,766 regular, 11 career and technical, 202 special program emphasis) [RECHECK/REVISE NUMBERS IF MODELING CHANGES].

Secondary Data. Analyses utilized several secondary data sources. The data on university characteristics come from the 2017 Integrated Postsecondary Education Data System (IPEDS) and 2020 Best Colleges ranking by U.S. News and World Report. Public high school measures are derived

[5]Web-scraping was conducted using the Python programming language using the *requests* library.

[6]For public universities in our analysis sample, we assessed the quality of the web-scraped data by asking the enrollment/admissions office for these data. If a university denied or ignored our request, we issued a formal public records request. Of the 16 public universities analyzed in this study, we received requested data from 11 universities. For these 11 universities, web-scraped data was very similar to publicly requested data. For universities that sent us data, the analyses below use "requested" rather than "scraped" data because requested data tended to have more events. Broad patterns were similar across requested data versus scraped data, and results based on scraped data are available upon request. Salazar et al. (2021) describe data quality checks in greater detail.

from the 2017-18 NCES Common Core of Data (CCD) and Niche 2020 Best Public High Schools ranking. Private high school measures are derived from 2017-19 NCES Private School Universe Survey (PSS) and Niche 2020 Best Private High Schools ranking.[7] ZIP code-level neighborhood characteristics were based on 5-year estimates from the 2020 American Community Survey (ACS), which includes data collected from 2016-2020. We defined neighborhood as the ZIP code the high school is located in.

5.2 Variables

Dependent variable. For RQ1, the dependent variable is counts of the number of visits from colleges to high schools during the 2017 calendar year. Here, if a college visited the same high school two times, this would count as two visits. RQ2 and RQ3 are addressed via regression analyses. Here, the dependent variable, $Y_{i,c}$, is a dichtomous indicator of whether high school i received one or more recruiting visits in 2017 from college c in our sample.

Independent variable. The independent variable of interest is College Board Geomarket. [CRYSTAL – REVISE]. We assigned each census tract to a College Board Geomarket, which we constructed through manual digitization of maps published by The College Board in 2023 marketing materials promoting their Enrollment Planning Service (EPS) software products (College Board 2023). Specifically, we georeferenced the College Board's Geomarket maps by overlaying PDF images onto spatial coordinates aligned with census tract boundaries. We then extracted vector data by manually tracing the Geomarket boundaries and generating individual shapefiles for each Geomarket. For instance, **?@fig-geomarket-methods** shows The College Board's map of Geomarkets in the Chicago metropolitan area on the left and our created shapefiles on the right. This approach to constructing Geomarkets aligned precisely with 2020 census tract boundaries from ACS 5-year estimates; that is, each census tract corresponded uniquely to a Geomarket generated from the College Board images. For example, **?@fig-geomarket-misalign** zooms into the northern Geomarkets of the Chicago metropolitan area from **?@fig-geomarket-methods**. The center map showcases how our created Geomarket shapefiles (delineated in purple) align with the spatial coordinates of 2020 census tract boundaries (delineated in grey).

[7] For both public and private high schools, we also used NCES data from previous years when the 2017-18 data was not available for a small number of schools.

FOR RQ1, we examine the relationship between student student market segment – local, in-state, regional, national – and high school visits. Following Zemsky, local visits are visits to high schools located in the same Geomarket as the college. In-state visits are visits to schools located in the same state, but a different Geomarekt. Regional visits are visits to schools located in the same EPS region, but a different state. National visits are to schools located in a different region. RQ1 also examines the relationship between EPS region – as defined by College Board – and high school visits. EPS regions are defined as follows: New England includes CT, ME, MA, NH, RI, and VT; Middle States includes NY, PA, DE, DC, MD, and NJ; Midwest includes IL, IN, IA, KS, MI, MN, MO, NE, ND, OH, SD, WV, and WI; South includes AL, FL, GA, KY, LA, MS, NC, SC, TN, and VA; Southwest includes AR, NM, OK, and TX; and West includes AK, AZ, CA, CO, HI, ID, MT, NV, OR, UT, WA, and WY. For RQ2, the primary independent variable is Geomarket but we also assess state and county as alternative predictors of high school recruiting visits.

Covariates. Regression analyses of the relationship between Geomarkets (X) and high school visits (Y) include a strong set of school-level covariates. We attempt to control for variables that plausibly affect whether a high school receives a visit from colleges in our sample. Decisions about which variables to include were motivated by prior research [e.g.,] (Stevens 2007; Salazar et al. 2021; Jaquette et al. 2024), logical argument, and data availability.

For the first set of RQ2 models, the analysis sample includes public and private high schools. Therefore, we control for variables that are available for both public and private schools. Prior research shows that private schools receive a disproportionate share of recruiting visits (Stevens 2007; Jaquette et al. 2024), despite enrolling fewer students on average than public schools. We include school control (public vs. private), grade 12 enrollment, and the interaction between these two variables. Prior research also finds that schools with a larger share of Black and Hispanic enrollment were less likely to receive visits (Salazar et al. 2021). We control for school-level enrollment shares in each of the following racial/ethnic categories, with percent White omitted as the reference category: Asian, Black, Hispanic, American Indian or Alaska Native (AI/AN), Native Hawaiian or Pacific Islander (NHPI), and two or more races. We control for school Niche "Best Schools" grade, which is treated as a categorical variable in regression models. Potential values are: A+, A, A-, B+, B, B-, C+, C, C-, and unranked. Models also control for several ZIP code-level "neighborhood" characteristics defined as the ZIP code the high school is located in: mean household income; percent of households who have earned a BA or higher; percent of households living below the poverty line; and percent of households by race/ethnicity (non-Hispanic Black, non-Hispanic Native American, non-Hispanic Asian, non-Hispanic Native Hawaiian or Pacific Islander, non-Hispanic multiracial, and Hispanic of any race). Prior research finds a non-linear relationship between neighborhood SES characteristics and high school visits (Salazar et al. 2021). Accordingly, the models operationalize household income and parental education as decile categories rather than as continuous measures. Finally, all models include a measure of the distance in miles between high school i and college j.

We also estimate separate models for public and private schools, which allows us to include covariates that are available for only one school sector. In addition to the covariates above, models of visits to public high schools include: percent of students receiving free/reduced lunch; the percent of students who pass the state math proficiency assessent; percent of students who pass the state reading/language arts assessment; and a dichotomous indicator for whether the school is a magnet school. Models of visits to private schools include a 5-category measure of religious affiliation (Nonsectarian, Catholic, Conservative Christian, Other Christian, Other Religion) and a dichotomous indicator of regular vs. "special program emphasis."

5.3 Analytic strategy

We address RQ1 using simple descriptive statistics that compare recruiting visit patterns to concepts from the Market Segment Model.

RQ2 asks whether Geomarkets explain which high schools receive recruiting visits, after controlling for other variables thought to explain recruiting. Equation 1 presents an OLS linear probability model of recruiting visits from a single college.

$$Y_i = \alpha + X_i\beta + \gamma_{g(i)} + \varepsilon_i \tag{1}$$

Subscript i indexes high schools. Y_i is a binary indicator equal to 1 if high school i receives a recruiting visit from the college under analysis and 0 otherwise. X_i is a vector of observed high school- and neighborhood-level covariates thought to shape recruiting. β is a vector of coefficients associated with the covariates in X_i. The term $\gamma_{g(i)}$ denotes a set of geographic fixed effects, where $g(i)$ identifies the geographic unit to which high school i belongs in a given specification. The term ε_i captures unexplained variation in whether high school i receives a recruiting visit, including omitted variables not captured by observed covariates and fixed effects.

The independent variable of interest is the Geomarket in which high school i is located. In a standard regression, a categorical variable is represented by coefficients on the non-reference categories, so the coefficients show how each category differs from the omitted reference category. That approach is not useful here because there are roughly 300 Geomarkets, and we do not want to estimate or present hundreds of Geomarket coefficients. More substantively, our question is not whether particular Geomarkets are positively or negatively associated with recruiting visits. Rather, the question is whether Geomarkets, as a broader classification system, help explain which high schools receive recruiting visits.

In Equation 1, $\gamma_{g(i)}$ denotes fixed effects for Geomarket. Conceptually, $\gamma_{g(i)}$ means that each Geomarket gets its own intercept. Therefore, the model allows the baseline probability of receiving a recruiting visit to differ across Geomarkets, consistent with the idea that schools in the same Geomarket share observed and unobserved factors that affect the likelihood of receiving a visit. Practically, Geomarket fixed effects are equivalent to including a dummy variable for each Geomarket, except that one Geomarket is omitted as the reference category. Geomarket fixed effects absorb all variation that differs between Geomarkets but is constant within Geomarkets. As a consequence, the coefficients in β are identified from variation between schools within the same Geomarket. Typically, fixed effects regression analyses are interested in estimating β after controlling for fixed effects. Our main interest is whether the inclusion of Geomarket fixed effects, $\gamma_{g(i)}$, improves the model's ability to explain recruiting visits, even after controlling for covariates, X_i. Therefore, we focus on measures of model fit. R^2 measures the share of variation in the dependent variable explained by the model. Adjusted R^2 modifies R^2 to penalize the inclusion of additional parameters that do not sufficiently improve model fit. We also estimate models with state and county fixed effects to compare their explanatory power to models with Geomarket fixed effects.

Whereas Equation 1 presents the model for a single college, Equation 2 presents the model of visits to school i by college j. Each observation is a high school-college pair, so the unit of analysis is a dyad. Each high school i therefore appears once for every college j in the sample. Outcome Y_{ij}, defined at the school-college level, equals 1 if high school i receives at least one recruiting visit from college j and 0 otherwise. X_i is a vector of observed high school- and neighborhood-level covariates that are rpeated across colleges for a given school, i. The term $\gamma_{j\times g(i)}$ denotes geographic fixed effects that are specific to each college. In other words, the model allows each college to have its own geographic pattern of recruiting. Practically, we estimate $\gamma_{j\times g(i)}$ by including fixed effects for the interaction between each college j and geography g.

$$Y_{ij} = \alpha + X_i\beta + \gamma_{j\times g(i)} + \varepsilon_{ij} \tag{2}$$

Outcomes, Y_{ij}, for the same school i are likely correlated across colleges, c, as are unobserved school traits. This data structure suggests correlation in ε_{ij} that violates the assumption of i.i.d. observations. We calcualate standard errors that are heteroskedasticity-robust and clustered two ways – by high school state and by college – to account for arbitrary correlation in outcomes within states (e.g., shared policy and recruiting environments) and within colleges (e.g., common admissions strategies and territory assignments across schools). This strategy allows for valid inference under these forms of dependence, but does not fully account for correlation across observations that share the same high school across colleges unless that dependence is captured by the state or college clustering dimensions.

Whereas the single college model in Equation 1 estimates how school characteristics relate to the probbability of a visit by a particular college, the Equation 2 model estimates how school characteristics relate to the probability of a visit across colleges. Once we include fixed effects for each combination of college and geography $\gamma_{j\times g(i)}$, coefficients are calculated based on variation between schools within the same college and geography. [IDEAS FOR THIS PARAGRAPH MAY GET MOVED]

We answer RQ2 by estimating a sequence of fixed effects linear probability models using the high school–college pair as the unit of analysis, following Equation 2. Appendix Tables X through X show results of models run separately by college, following Equation 1. Across specifications, we vary the geographic structure absorbed by the fixed effects, beginning with college fixed effects only and then adding college-specific fixed effects for state, county, and Geomarket. We estimate each set of models both without and with observed school- and neighborhood-level covariates. The central question is whether models that absorb college-by-Geomarket differences explain recruiting visits better than models that absorb coarser geographic units. Because the aim is to assess the explanatory power of Geomarkets as a classification system rather than the effect of any particular Geomarket, we focus on model fit statistics, especially adjusted R^2

RQ3 asks whether the relationship between Geomarket popularity and the probability of receiving a visit varies by school socioeconomic and racial composition. Because the goal of RQ3 is to assess how the relationship between school composition and recruiting varies with Geomarket context, we focus on the estimated coefficients rather than model fit. All RQ3 models include fixed effects for the interaction between state and college. Therefore, coefficients are identified based on variation between high schools within the same college and state. The coefficient on Geomarket popularity —- measured as the average number of visits per school —- captures how the probability that a high school receives a visit varies across schools located in Geomarkets with different levels of recruiting intensity, within the same college and state. The coefficients on school composition capture differences in the probability of receiving a visit across schools with different socioeconomic and racial profiles. The key quantities of interest are the interaction terms between Geomarket popularity and school composition. These coefficients indicate whether the association between Geomarket popularity and the probability of a visit differs across schools with different socioeconomic or racial compositions. Substantively, they show whether increases in Geomarket popularity are associated with larger gains in recruiting for some types of schools than for others. [WHAT SHOULD I SAY ABOUT ESTIMATES NOT BEING CAUSAL?]

6 Results

6.1 Research Question 1

RQ1 asks, to what extent are visit patterns consistent with recommendations by the Market Segment Model? We argue that, for private institutions and most public institutions in our sample, the Market Segment Model suggests focusing on affluent Geomarkets in regions where many students are likely to consider a college like yours.

Appendix Figure X provides context by showing freshman enrollment by residency status – in-state, domestic out-of-state, and international – for the colleges in our sample. International students comprise a small share of freshman for most colleges in our sample, but approaches 20% for several institutions (e.g., Smith, MaCalaster, Emory, UC-Irvine, and UC-San Diego).

In Figure X, the distribution of high school recruiting visits across market segments—local/in-state, regional, and national—shows that colleges, especially selective private institutions, organize their search for students around regional and national markets. Among selective private colleges, recruiting visits are concentrated in the regional and national segments, which is consistent with the Market Segment Model's claim that selective colleges primarily draw students from markets beyond the local area. At public universities, recruiting is more varied. Many distribute visits across in-state, regional, and national markets, and several undertake substantial out-of-state recruiting. University of Alabama (3,629 visits) and CU Boulder (1,283 visits) stand out for especially strong national recruiting footprints, while UC Irvine (499), UC San Diego (999), and UC Riverside (629) are clearer examples of in-state recruiting strategies [CRYSTAL – REVISE VISIT NUMBERS].

Appendix Figure X compares the distribution of recruiting visits and domestic freshman enrollment across the same market segments. The comparison suggests that recruiting behavior is more consistent with the more expansive regional and national orientation of the the Market Segment Model, while freshman enrollment remains more constrained by geography. This contrast is especially visible among public universities: many recruit across in-state, regional, and national markets, even when enrollment remains heavily concentrated in-state. Among selective private colleges, both recruiting and enrollment are concentrated in the regional and national segments, but recruiting is often more national than enrollment.

In Figure X, the distribution of recruiting visits across EPS regions—New England, Middle States, Midwest, South, Southwest, and West—shows that colleges organize recruiting within a broad geography of regional proximity, but with substantial variation in how tightly recruiting is anchored to the home region and home state. For most colleges, especially public universities, recruiting is concentrated in the home region and adjacent regions, with more distant regions receiving smaller shares of visits. Within the home region, however, colleges vary in how tightly recruiting is anchored to the home state. The horizontal line within each home-region cell indicates the in-state share of recruiting within that region. This contrast is especially visible among public universities: Alabama, South Carolina, and CU Boulder devote substantial recruiting effort to their home region, but much of that home-region recruiting is directed to high schools outside the home state, whereas the UC campuses recruit heavily within the West and direct the overwhelming majority of that home-region recruiting to California high schools. Selective private colleges generally show a broader geographic recruiting footprint across regions, which is consistent with the Market Segment Model's claim that more selective institutions draw students from wider markets. Overall, Figure X suggests that geographic region strongly structures recruiting, with most colleges concentrating visits within their home region or their home region plus geographically adjacent regions.

Appendix Figure X compares the distribution of recruiting visits and domestic freshman enrollment across the same EPS regions. The clearest pattern is that enrollment remains more regionally concentrated than recruiting. For most colleges, especially public universities, freshman enrollment follows a geography of adjacency: the largest share comes from the home region, followed by adjacent regions, with distant regions contributing much smaller shares. The comparison also shows that home-region enrollment is often more tightly anchored to the home state than home-region recruiting. Selective private colleges show a broader geographic footprint in both recruiting and enrollment, which is consistent with the Market Segment Model's claim that more selective institutions draw students from wider markets. Overall, recruiting visits thus follow a similar geography as enrollment but are usually more dispersed, indicating that colleges search for students across a broader set of regions than those from which they ultimately enroll freshmen.

Tables X shows the top 30 Geomarkets ranked by the number of visits per public high school. Table X shows the same table, but focusing on visits to private high schools. We separate visits to public and private high schools because the relationship between Geomarkets and high school visits differs substantially for public versus private schools. The top row shows mean values, treating each Geomarket equally, regardless of size. For public high school visits in Table X, the top 30 most popular Geomarkets are consistently affluent, with average household income well North of $105k average. These Geomarkets also tend to have a higher proportion of BA completers and a lower proportion of households in poverty. The top 30 geomarkets skew toward higher shares of white and Asian residents relative to the 62.7% and 5.6% averages across all geomarkets, while Black and Hispanic shares tend to be lower — consistent with the socioeconomic patterns noted above, though with notable exceptions such as GA 2 - Fulton Co (%Black = 43.1) and CA26 - Santa Ana (%Hisp = 38.9).

Table X presents results for private schools. Table X presents results for private schools. The Geomarekt affluence gradient [use different term] visible for visits to public schools is much weaker for visits to private schools. Several of the highest-ranked geomarkets — TX19 - City of Dallas ($81k, ranked 2nd), TX20 - City of Fort Worth ($91k, 4th), and MO 1 - Kansas City ($84k, 23rd) — have household incomes below the all-geomarket mean, yet rank among the most intensively visited for private schools. This reflects the fact that elite private schools concentrate in urban cores regardless of surrounding neighborhood income. Moreover, these results raise the possibility that Geomarkets may be less central to private school recruiting efforts than they are to public school recruiting efforts.

Table X presents results for visits to public schools in the national market segment, defined as public schools located in a different EPS region than the college.[1] Because in-state and regional visits are excluded, visits per school are lower than in Table X overall. The composition of the top 30, however, shifts toward wealthier geomarkets. CA11 - Santa Clara Co ($234k) rises from 11th to 4th, and CA 9 - San Mateo Co ($196k) enters the top 30 for the first time at 26th. Meanwhile, CA26 - Santa Ana ($131k, ranked 22nd overall) and NJ 7 - Union Co ($114k, was #27) drop entirely off the national list entirely. The pattern suggests that when universities recruit beyond their home region, they concentrate even more heavily on high-income geomarkets than their overall visit patterns imply.

Figure X plots Lorenz concentration curves, familiar from analyses of income inequality. Each point on a curve represents a Geomarket. The Y-axis is the cumulative share of visits; the X-axis is the cumulative share of schools, ranked in ascending order by mean household income at the Geomarket level. The dashed 45-degree line represents parity, where the bottom half of schools would receive exactly half of all visits. The Affluence Tilt Index (ATI) measures the area between the curve and the diagonal, rescaled to range from -1 to 1, and summarizes each curve as a single number comparable across institution types and geographic scopes. An ATI of zero means visits are perfectly proportional to the distribution of schools; positive values indicate that visits are concentrated among higher-income geomarkets, with larger values reflecting greater tilt toward affluence. The red line shows that across all visits, the bottom half of schools received only about 28% of visits — meaning the top half received roughly 72% — yielding an ATI of 0.33. The intersection of the horizontal reference line (Y = 0.5) with the red line further illustrates this: half of all visits were concentrated in the top 25% of schools by geomarket income. Visits to private schools (green, ATI = 0.15) show a substantially weaker affluence gradient than visits to public schools (blue, ATI = 0.37). That is, colleges made relatively more visits to private schools in lower-income geomarkets compared to public schools in similarly low-income geomarkets.

Figure Y presents concentration curves in a 3×3 grid, with columns for institution type (private research, private liberal arts, public research) and rows for high school market segment (in-state, in-region, national). Beginning with the top row, public research universities made 3,870 in-state visits – about a quarter of their 16,052 total – and their curves sit close to the diagonal, reflecting near-proportional recruiting across the Geomarket income distribution. The private research and private liberal arts panels tell a different story: even for in-state visits, curves bow noticeably below the diagonal, indicating that these institutions tilt toward affluent Geomarkets even when recruiting close to home. Moving down any column to the in-region and national rows, curves pull progressively further below the diagonal across all institution types. Within each cell, visits to public high schools (blue) consistently fall further below the diagonal than visits to private high schools (green). This pattern culminates in the national row, where public school visits by public research universities exhibit the highest affluence tilt in the entire grid: roughly 60% of such visits are concentrated in the top fifth of Geomarkets by income, yielding an ATI of 0.51, compared to 0.47 and 0.42 for private research and private liberal arts universities, respectively.

6.2 Research Question 2

RQ2 asks whether Geomarkets explain which high schools receive recruiting visits, net of factors thought to affect recruiting. Table X compares fixed-effects linear probability models of whether high school i receives a recruiting visit from college j. The outcome is Y_{ij}, an indicator equal to 1 if high school i receives a visit from college j. Across specifications, the models vary in how they structure college-specific geography, allowing us to compare the explanatory power of models with Geomarket fixed effects to models with alternative geographic specifications. Column 1 includes fixed effects for colleges. Column 2 includes fixed effects for the interaction between college and state. Column 3 includes fixed effects for the interaction between college and county. Column 4 includes fixed effects for the interaction between college and Geomarket. Models (1) through (4), shown in the top panel, do not include covariates. Models (5) through (8), shown in the bottom panel, include the following covariates: school sector interacted with 12th-grade enrollment, school racial/ethnic composition, overall Niche letter grade, ZIP code-level mean household income decile, ZIP code-level share of adults with a bachelor's degree or higher decile, ZIP code-level poverty rate (and its squared term), ZIP code-level racial/ethnic composition, and distance from the high school to the college. Regression coefficients for all variables are shown in Appendix Table X. The main purpose of Table X is to assess how well Geomarkets capture the territorial structure of recruiting visits relative to alternative geographic specifications. Because the state, county, and Geomarket models differ substantially in dimensionality, we focus primarily on adjusted R^2, the proportion of variation explained by the model, adjusted downward to penalize the inclusion of parameters that do not sufficiently improve model fit.

The all-schools results provide strong support for the argument that Geomarkets capture meaningful territorial structure in recruiting behavior. In the models without covariates, adjusted R^2 increases from .009 in the university fixed-effects specification to .107 in the university-by-state specification and .186 in the university-by-Geomarket specification, indicating that Geomarkets explain substantially more variation in recruiting visits than states alone. County models achieve higher unadjusted R^2 than Geomarket models, which is unsurprising given that counties provide a much finer-grained partition of space. Model (3), which includes fixed effects for the interaction between college and county, contains 127,471 parameters ($= 3{,}035$ counties$\times 42$ colleges$+1$). However, the county models perform substantially worse than the Geomarket models on adjusted R^2 (.116 versus .186), indicating that Geomarkets capture the territorial structure of recruiting visits more parsimoniously. The same pattern holds after adding covariates: adjusted R^2 rises from .139 in the university fixed-effects model to .219 in the university-by-state model and .272 in the university-by-Geomarket model, while the county specification reaches only .209. Taken together, these results suggest that Geomarkets are not simply coarser substitutes for conventional geographic units, but a more efficient organizational geography for structuring recruiting visits. Interpreted through the lens of performativity, the Geomarkets have more explanatory power than alternative geographic specifications precisely because they are an important component of the organizational and vendor infrastructure that produces recruiting visits: colleges often assign admissions officers to territories defined by Geomarkets; College Board EPS software encouraged colleges to identify desirable Geomarkets as recruiting markets; and Slate CRM software integrates Geomarkets into functionality for recruitment planning and assessment.

The separate models for public schools (Table X) and private schools (Table X) reinforce the main pattern observed in the all-schools models. In the public-school models without covariates, adjusted R^2 increases from .134 in the university-by-state specification to .158 in the university-by-county specification and .236 in the university-by-Geomarket specification. The bottom panel adds the covariates included in the all-schools models as well as four measures available only for public schools: percent of students receiving free or reduced-price lunch, math proficiency, reading/language arts proficiency, and an indicator for magnet school status. The estimation sample declines from 17,503 to 16,160 schools because these covariates – especially the proficiency measures – are missing for some schools. Even with the smaller sample, the same pattern holds: adjusted R^2 rises to .233 in the university-by-state model, .241 in the university-by-county model, and .312 in the university-by-Geomarket model. As in the all-schools sample, county models achieve higher unadjusted R^2 than Geomarket models but lower adjusted R^2, indicating that Geomarkets capture the territorial structure of public-school recruiting visits more parsimoniously than either states or counties. Results for the private-school sample, shown in Table X, are consistent with the public-school results, with two exceptions. First, university-by-county models perform especially poorly on adjusted R^2 in the private-school sample, suggesting that counties are a particularly inefficient geography for structuring recruiting to private schools. Second, the gain in adjusted R^2 from replacing university-by-state fixed effects with university-by-Geomarket fixed effects is larger in the public-school models than in the private-school models: without covariates, the increase is .102 for public schools (.134 to .236) but .056 for private schools (.098 to .154), and with covariates the increase is .079 for public schools

(.233 to .312) but .038 for private schools (.232 to .270). Thus, Geomarkets add explanatory power in both sectors, but they explain substantially more variation in recruiting visits to public schools than to private schools.

Robustness tests. As a robustness test, we re-estimate the RQ2 models after restricting the analysis sample to high schools located close to a Geomarket boundary. The logic is compare schools that are close to one another while preserving the same basic modeling framework: if the explanatory power of Geomarkets in the full-sample models simply reflects broad differences across distant places, then the advantage of university-by-Geomarket fixed effects should weaken when the sample is limited to schools near boundaries. At the same time, this design should not be interpreted as a regression discontinuity because we are not estimating a discontinuous change in visit probability at a particular border and it should not be interpreted as a boundary-discontinuity analysis because we are not comparing schools on opposite sides of the same border segment. Instead, we examine whether Geomarket fixed effects continue to improve model fit in a restricted sample of border-proximate schools. This restricted-sample approach is therefore best understood as a practical and more demanding robustness test of the original fixed-effects models: it moves the comparison closer to Geomarket boundaries without abandoning the original estimand or sacrificing as much statistical power as a true border-pair design would.

Table X reports robustness models that compare two geographic specifications: fixed effects for the interaction between university and state, and fixed effects for the interaction between university and Geomarket. For each specification, we re-estimate the models after restricting the sample to high schools located within 2 miles, 1 mile, and 0.5 miles of a Geomarket boundary. The substantive pattern from the main RQ2 models persists under each boundary restriction. In the models without covariates, adjusted R^2 remains higher for the university-by-Geomarket models than for the university-by-state models among schools within 2 miles of a boundary (.163 versus .097), within 1 mile (.159 versus .077), and within 0.5 miles (.165 versus .077). The same is true once covariates are added: adjusted R^2 is .257 versus .215 within 2 miles, .259 versus .202 within 1 mile, and .270 versus .211 within 0.5 miles. Thus, the explanatory advantage of university-by-Geomarket fixed effects survives increasingly stringent boundary restrictions, suggesting that the main RQ2 results are not driven only by broad spatial differences across distant places.

The public-school boundary models, shown in Table X, point in the same direction. Under each boundary restriction, university-by-Geomarket fixed effects continue to outperform university-by-state fixed effects on adjusted R^2, both without covariates and with covariates. Models for the sample of private schools, shown in Appendix Table X, show the same patterns. Thus, the main RQ2 pattern remains visible in both sectors even when the analysis is limited to schools located close to Geomarket boundaries.

6.3 Research Question 3

7 Discussion

8 References

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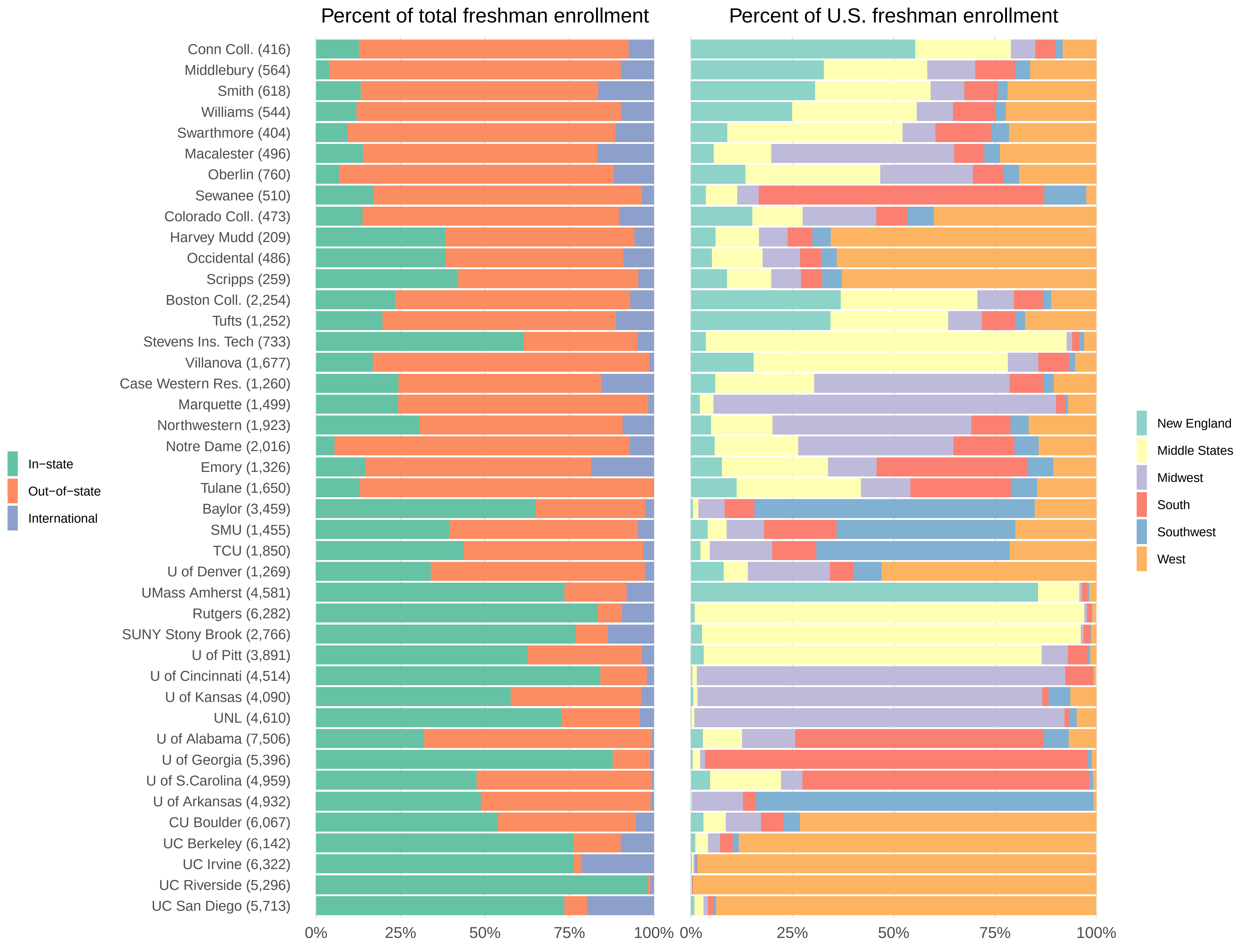
Percent of total freshman enrollment
Percent of U.S. freshman enrollment
Conn Coll. (416)
Middlebury (564)
Smith (618)
Williams (544)
Swarthmore (404)
Macalester (496)
Oberlin (760)
Sewanee (510)
Colorado Coll. (473)
Harvey Mudd (209)
Occidental (486)
Scripps (259)
Boston Coll. (2,254)
Tufts (1,252)
Stevens Ins. Tech (733)
Villanova (1,677)
Case Western Res. (1,260)
Marquette (1,499)
Northwestern (1,923)
Notre Dame (2,016)
Emory (1,326)
Tulane (1,650)
Baylor (3,459)
SMU (1,455)
TCU (1,850)
U of Denver (1,269)
UMass Amherst (4,581)
Rutgers (6,282)
SUNY Stony Brook (2,766)
U of Pitt (3,891)
U of Cincinnati (4,514)
U of Kansas (4,090)
UNL (4,610)
U of Alabama (7,506)
U of Georgia (5,396)
U of S.Carolina (4,959)
U of Arkansas (4,932)
CU Boulder (6,067)
UC Berkeley (6,142)
UC Irvine (6,322)
UC Riverside (5,296)
UC San Diego (5,713)
0%
25%
50%
75%
100%
In−state
Out−of−state
International
New England
Middle States
Midwest
South
Southwest
West

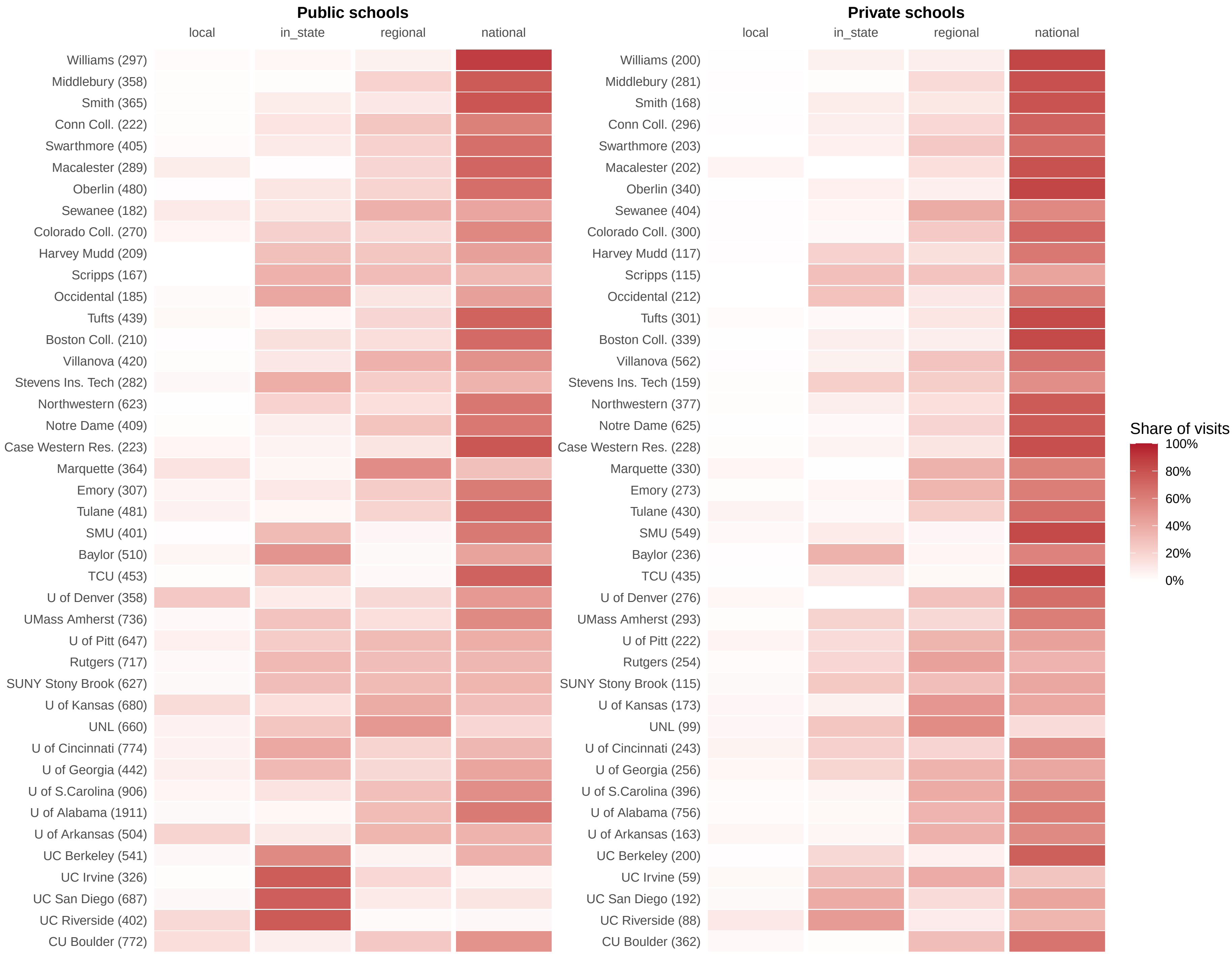

Figure 2: Recruiting visits by market segment of Geomarket

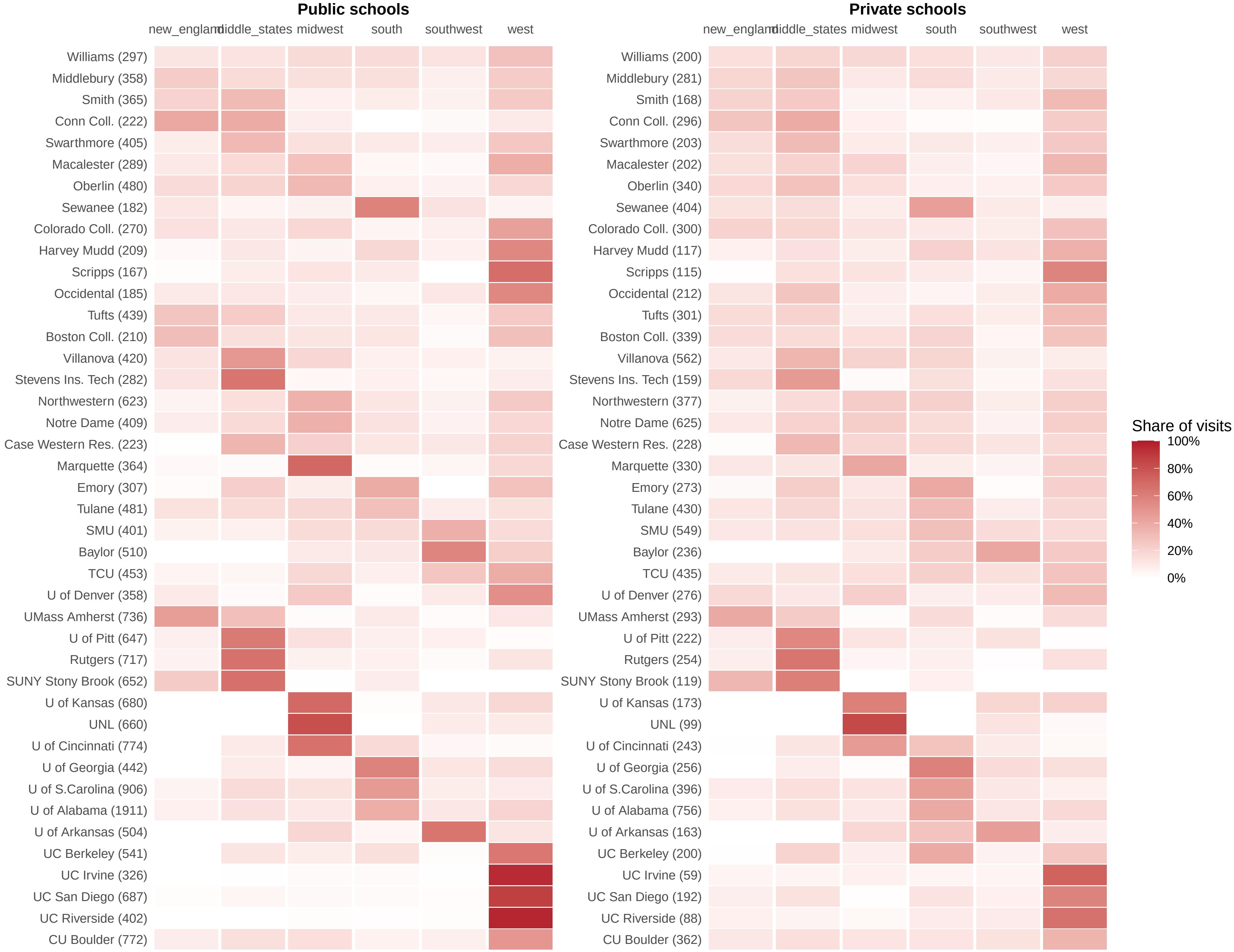

Figure 3: Recruiting visits by EPS region

Table 1: Top 30 Geomarkets ranked by visits per school (public school visits only)

Rank	EPS	Visits/Sch	Schools	Visits	MeanInc	%BA+	%Pov	%White	%Asian	%Black	%Hisp
—	Mean (All Geomarkets)	2.2	33	74	$105k	33.5	12.2	62.7	5.6	11.4	16.4
1	IL 9 - North Shore	24.5	6	147	$259k	73.5	5.6	79.0	9.6	1.9	7.2
2	IL10 - Evanston & Skokie	17.0	5	85	$156k	58.3	8.4	64.6	17.7	5.1	9.4
3	VA 1 - Arlington & Alexandria	11.6	5	58	$175k	71.4	6.6	57.1	8.8	13.9	15.8
4	TX23 - Collin & Rockwall Co	11.2	25	281	$146k	52.3	6.0	56.8	14.5	9.4	15.6
5	IL 8 - Northwest Suburbs	11.0	13	143	$126k	44.7	6.8	58.2	15.0	3.3	20.8
6	MD 2 - Montgomery Metropolitan	10.4	25	260	$165k	59.2	6.6	43.1	14.9	18.0	19.5
7	VA 2 - Fairfax Co	10.3	26	269	$182k	62.3	5.4	50.4	19.6	9.3	16.2
8	IL12 - Western Suburbs	10.3	35	359	$134k	46.9	6.8	60.3	10.5	8.1	18.5
9	CT 3 - Fairfield Co	9.9	31	308	$160k	48.9	8.7	61.0	5.3	10.6	20.0
10	IL 7 - Chain of Lakes	9.7	15	145	$125k	39.7	8.4	57.2	7.4	7.5	25.1
11	CA11 - Santa Clara Co (excluding San Jose)	9.6	19	182	$234k	64.3	5.7	36.8	39.4	1.8	17.1
12	CO 2 - Metro Denver & Northeastern Colorado	9.4	103	966	$121k	44.8	8.7	66.2	3.8	4.3	22.1
13	CA 4 - Marin Co	9.0	7	63	$190k	60.2	6.6	70.6	5.7	2.1	16.1
14	NY17 - Northern Nassau Co	8.8	14	123	$266k	62.4	6.7	67.7	16.9	2.3	10.6
15	NJ 6 - Somerset & Mercer Co	8.8	22	193	$155k	49.3	8.5	51.8	14.6	14.8	16.4
16	CA28 - South Orange Co	8.2	14	115	$177k	56.7	7.2	57.4	19.5	1.4	16.5
17	CA17 - West Los Angeles & West Beach	8.2	11	90	$155k	60.3	11.1	50.3	11.9	9.6	22.5
18	MA10 - Milton, Lexington, & Waltham	8.1	9	73	$181k	64.0	5.8	74.2	13.1	3.4	6.0
19	GA 2 - Fulton Co	7.7	25	193	$116k	54.5	12.3	39.3	7.3	43.1	7.2
20	NY19 - Northwest Suffolk Co	7.7	16	123	$182k	47.7	4.6	76.9	5.9	3.0	12.0
21	CA 6 - Contra Costa Co	7.2	29	209	$151k	43.3	8.1	42.6	17.2	8.2	25.8
22	CA26 - Santa Ana	7.0	28	195	$131k	36.5	10.4	35.1	21.1	1.1	38.9
23	TX24 - West of Dallas/Ft. Worth Metroplex	6.9	38	264	$124k	41.3	6.9	62.6	6.9	8.9	17.9
24	NJ11 - Morris & Northern Passaic Co	6.8	25	169	$173k	53.4	5.3	71.8	9.7	2.9	13.2
25	NJ10 - Bergen Co	6.7	39	263	$159k	50.7	7.2	55.5	16.4	5.3	20.4
26	CA 8 - Alameda Co (w/o Oakland)	6.7	26	173	$165k	49.3	7.6	31.0	36.4	6.0	20.8
27	NJ 7 - Union Co	6.6	22	145	$114k	37.1	8.9	39.0	5.2	20.2	32.0
28	PA 4 - Montgomery Co	6.6	21	138	$142k	49.6	6.2	74.9	7.7	9.1	5.3
29	CA20 - South Bay	6.5	21	137	$126k	44.6	8.9	31.7	21.8	9.8	31.7
30	MA 8 - Lowell, Concord, & Wellesley	6.5	28	181	$160k	51.6	7.2	75.0	10.7	3.4	7.4

Table 2: Top 30 Geomarkets ranked by visits per school (private school visits only)

Rank	EPS	Visits/Sch	Schools	Visits	MeanInc	%BA+	%Pov	%White	%Asian	%Black	%Hisp
—	Mean (All Geomarkets)	6.6	6	40	$105k	33.5	12.2	62.7	5.6	11.4	16.4
1	IL 9 - North Shore	31.0	1	31	$259k	73.5	5.6	79.0	9.6	1.9	7.2
2	TX19 - City of Dallas	22.4	12	269	$81k	34.9	15.6	29.8	3.4	22.9	41.7
3	IL10 - Evanston & Skokie	20.3	3	61	$156k	58.3	8.4	64.6	17.7	5.1	9.4
4	TX20 - City of Fort Worth	20.2	5	101	$91k	28.3	12.1	42.0	4.4	15.8	34.8
5	CA 7 - City of Oakland	19.5	4	78	$124k	47.0	13.4	29.6	15.7	21.7	26.3
6	IL 7 - Chain of Lakes	19.0	1	19	$125k	39.7	8.4	57.2	7.4	7.5	25.1
7	VA 1 - Arlington & Alexandria	18.8	4	75	$175k	71.4	6.6	57.1	8.8	13.9	15.8
8	CA 4 - Marin Co	17.7	3	53	$190k	60.2	6.6	70.6	5.7	2.1	16.1
9	DC 1 - District of Columbia	17.0	10	170	$155k	59.8	13.7	36.7	4.0	44.5	11.1
10	TX16 - Southwest Houston Metro Area	16.6	13	216	$106k	46.8	11.8	31.5	16.1	20.0	29.6
11	CA29 - North San Diego Co (w/o San Diego)	16.0	4	64	$144k	44.1	8.0	52.3	10.4	2.3	29.8
12	CT 4 - Waterbury & Litchfield Co	16.0	6	96	$129k	35.9	7.2	87.6	1.9	1.5	6.6
13	NH 1 - Seacoast	16.0	2	32	$138k	44.5	7.1	91.2	3.0	0.9	2.3
14	GA 2 - Fulton Co	14.9	15	224	$116k	54.5	12.3	39.3	7.3	43.1	7.2
15	CA28 - South Orange Co	14.2	8	114	$177k	56.7	7.2	57.4	19.5	1.4	16.5
16	CT 5 - Hartford & Tolland Co	14.1	9	127	$117k	39.0	11.0	63.5	5.4	11.4	16.6
17	CA31 - City of San Diego	14.0	5	70	$118k	45.4	11.4	44.7	14.7	6.8	28.9
18	MA 7 - Quincy & Plymouth Co	13.2	4	53	$137k	38.6	7.5	80.2	1.4	8.9	4.0
19	TX 6 - Austin & Central Texas	13.1	9	118	$114k	43.0	9.4	53.7	5.0	6.4	31.9
20	NY19 - Northwest Suffolk Co	13.0	2	26	$182k	47.7	4.6	76.9	5.9	3.0	12.0
21	OH 5 - Cuyahoga, Geauga, & Lake Co	12.8	5	64	$106k	36.4	7.1	87.2	2.3	4.6	3.5
22	OR 1 - Greater Portland (West)	12.8	8	102	$130k	53.6	9.4	69.6	9.0	3.3	12.1
23	MO 1 - Kansas City & St. Joseph	12.6	9	113	$84k	28.7	11.9	76.0	1.5	11.9	6.7
24	NJ 6 - Somerset & Mercer Co	12.2	11	134	$155k	49.3	8.5	51.8	14.6	14.8	16.4
25	CA15 - San Fernando Valley (East)	12.1	8	97	$123k	51.6	12.0	58.4	6.8	5.1	24.9
26	CO 2 - Metro Denver & Northeastern Colorado	12.0	14	168	$121k	44.8	8.7	66.2	3.8	4.3	22.1
27	MD 2 - Montgomery Metropolitan	11.9	15	179	$165k	59.2	6.6	43.1	14.9	18.0	19.5
28	CA11 - Santa Clara Co (excluding San Jose)	11.6	5	58	$234k	64.3	5.7	36.8	39.4	1.8	17.1
29	CA26 - Santa Ana	11.5	2	23	$131k	36.5	10.4	35.1	21.1	1.1	38.9
30	CT 2 - New Haven & Middlesex Co	11.3	12	136	$118k	37.1	10.8	65.3	3.9	11.3	16.6

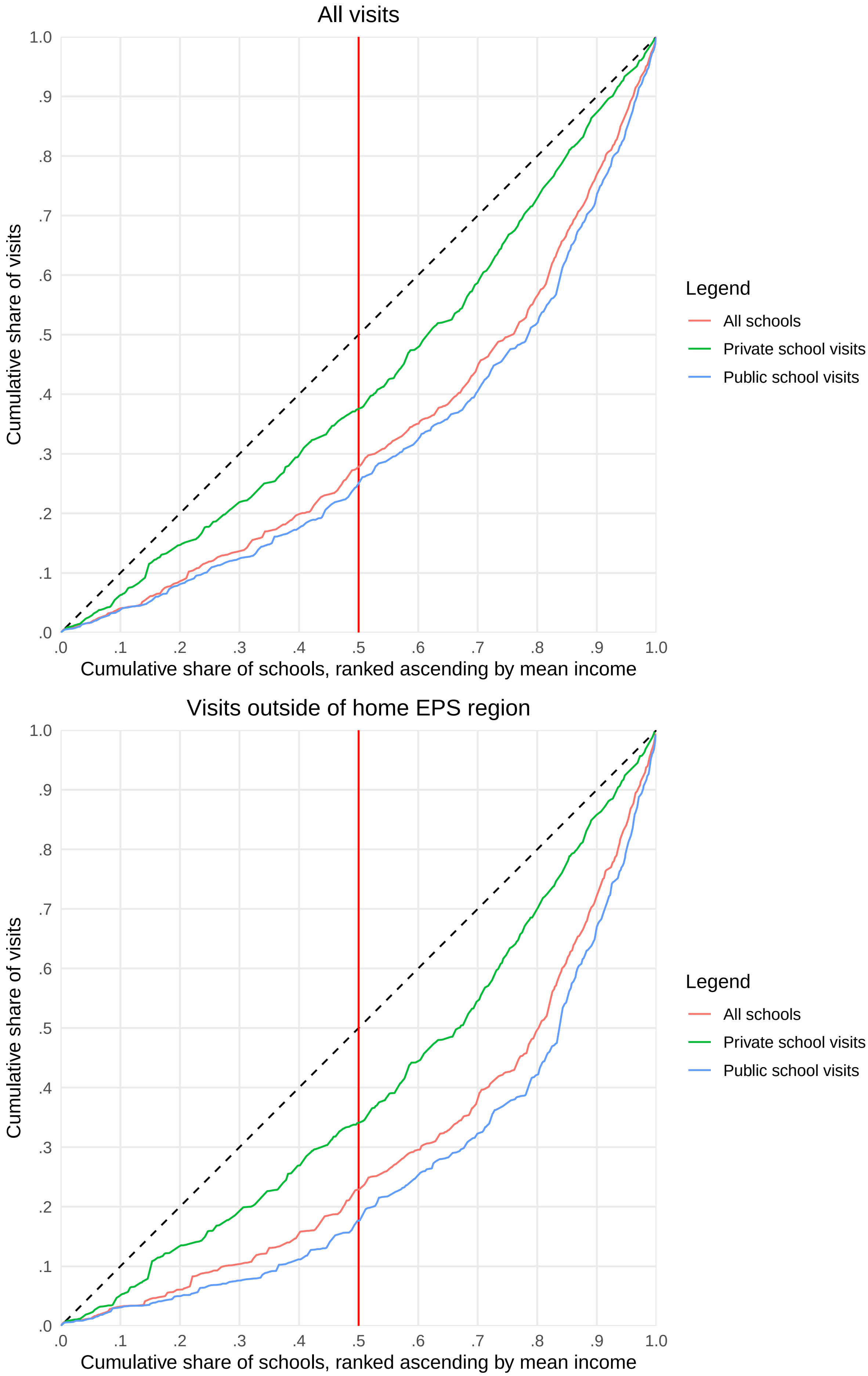

Figure 4: Lorenz-style concentration curves of recruiting visits by Geomarket affluence

Table 3: Model Fit Statistics, fixed effects regression of visit to high school i by college j, separate models for all HS, public HS, private HS

term	(1) Univ FE	(2) Univ × State FE	(3) Univ × County FE	(4) Univ × EPS FE	(5) Covar + Univ FE	(6) Covar + Univ × State FE	(7) C
All High Schools							
R^2	0.009	0.109	0.242	0.197	0.139	0.221	0.323
Adj. R^2	0.009	0.107	0.116	0.186	0.139	0.219	0.209
Within R^2	0.000	0.000	0.000	0.000	0.131	0.125	0.107
RMSE	0.182	0.172	0.159	0.164	0.170	0.161	0.150
Public High Schools Only							
N	NA	NA	NA	NA	NA	NA	NA
Adj. R^2							
Within R^2							
RMSE							
Private High Schools Only							
N	NA	NA	NA	NA	NA	NA	NA
Adj. R^2							
Within R^2							
RMSE							

Table 4: Fixed Effects Regression Model with Interaction between Geomarket Popularity and School SES/Racial Composition

	Model		Interaction Free-Reduced Lunch		Interaction Black/Hisp	
	Estimate	Std Error	Estimate	Std Error	Estimate	Std Error
hs_pct_free_reduced_lunch_decileD2	-0.008*	0.003	0.016***	0.003	-0.007*	0.003
hs_pct_free_reduced_lunch_decileD3	-0.006	0.004	0.025***	0.004	-0.004	0.003
hs_pct_free_reduced_lunch_decileD4	-0.005	0.004	0.030***	0.004	-0.003	0.004
hs_pct_free_reduced_lunch_decileD5	-0.003	0.004	0.034***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD6	-0.004	0.004	0.033***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD7	-0.004	0.004	0.035***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD8	-0.002	0.004	0.036***	0.005	0.001	0.004
hs_pct_free_reduced_lunch_decileD9	-0.001	0.005	0.038***	0.005	0.002	0.004
hs_pct_free_reduced_lunch_decileD10	-0.000	0.004	0.043***	0.004	0.002	0.004
hs_pct_bl_hisp_nat_decileD1	0.005	0.002	0.006*	0.002	0.020***	0.003
hs_pct_bl_hisp_nat_decileD2	0.003	0.002	0.004	0.002	0.012***	0.003
hs_pct_bl_hisp_nat_decileD3	0.003	0.003	0.003	0.002	0.006*	0.002
hs_pct_bl_hisp_nat_decileD4	0.002	0.002	0.001	0.002	0.002	0.002
hs_pct_bl_hisp_nat_decileD6	-0.005***	0.001	-0.004**	0.001	-0.002	0.002
hs_pct_bl_hisp_nat_decileD7	-0.004*	0.002	-0.001	0.002	0.001	0.003
hs_pct_bl_hisp_nat_decileD8	-0.007*	0.003	-0.002	0.002	0.004	0.003
hs_pct_bl_hisp_nat_decileD9	-0.011*	0.004	-0.003	0.004	0.006	0.003
hs_pct_bl_hisp_nat_decileD10	-0.016**	0.005	-0.005	0.005	0.012**	0.004
peps_n_vis01_per_sch_all	0.504***	0.035	0.955***	0.049	0.695***	0.046
hs_pct_free_reduced_lunch_decileD2:peps_n_vis01_per_sch_all			-0.327***	0.032		
hs_pct_free_reduced_lunch_decileD3:peps_n_vis01_per_sch_all			-0.480***	0.041		
hs_pct_free_reduced_lunch_decileD4:peps_n_vis01_per_sch_all			-0.569***	0.047		
hs_pct_free_reduced_lunch_decileD5:peps_n_vis01_per_sch_all			-0.629***	0.048		
hs_pct_free_reduced_lunch_decileD6:peps_n_vis01_per_sch_all			-0.609***	0.047		
hs_pct_free_reduced_lunch_decileD7:peps_n_vis01_per_sch_all			-0.675***	0.043		
hs_pct_free_reduced_lunch_decileD8:peps_n_vis01_per_sch_all			-0.665***	0.059		
hs_pct_free_reduced_lunch_decileD9:peps_n_vis01_per_sch_all			-0.697***	0.071		
hs_pct_free_reduced_lunch_decileD10:peps_n_vis01_per_sch_all			-0.813***	0.039		
hs_pct_bl_hisp_nat_decileD1:peps_n_vis01_per_sch_all					-0.456***	0.067
hs_pct_bl_hisp_nat_decileD2:peps_n_vis01_per_sch_all					-0.200**	0.058
hs_pct_bl_hisp_nat_decileD3:peps_n_vis01_per_sch_all					-0.067	0.059
hs_pct_bl_hisp_nat_decileD4:peps_n_vis01_per_sch_all					-0.005	0.040
hs_pct_bl_hisp_nat_decileD6:peps_n_vis01_per_sch_all					-0.089**	0.028
hs_pct_bl_hisp_nat_decileD7:peps_n_vis01_per_sch_all					-0.128***	0.035
hs_pct_bl_hisp_nat_decileD8:peps_n_vis01_per_sch_all					-0.223***	0.042
hs_pct_bl_hisp_nat_decileD9:peps_n_vis01_per_sch_all					-0.316***	0.052
hs_pct_bl_hisp_nat_decileD10:peps_n_vis01_per_sch_all					-0.487***	0.058
N	678,804		678,804		678,804	
Adj. R^2	0.272		0.298		0.282	
Within R^2	0.151		0.181		0.162	
RMSE	0.140		0.138		0.139	

Significance codes: *** p<0.001, ** p<0.01, * p<0.05

9 Appendix